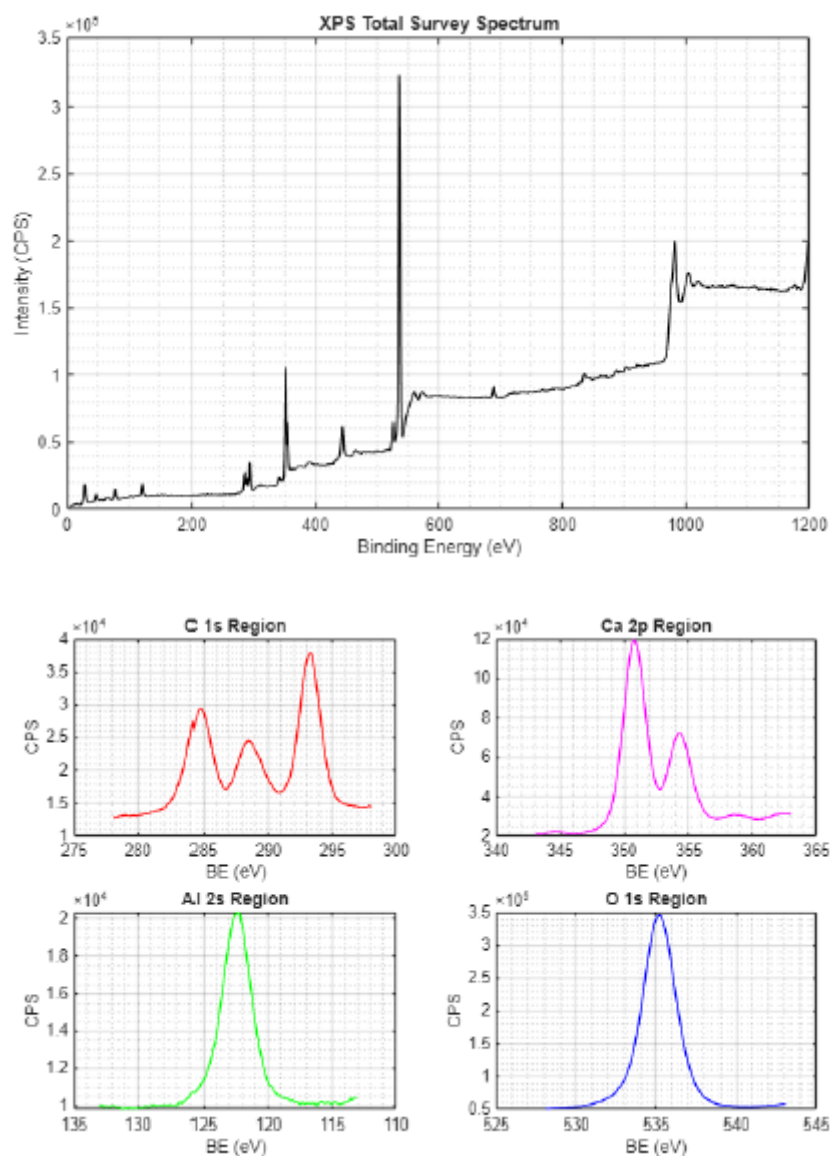


XPS Laboratory Guidelines and Report

The experimental problem proposed in this laboratory session has been fully solved and compiled. The comprehensive quantitative results, tracking profiles, and final plots can be found in the accompanying PDF document generated directly via the MATLAB `publish` routine.

1. Data Initialization and Preliminary Visualization

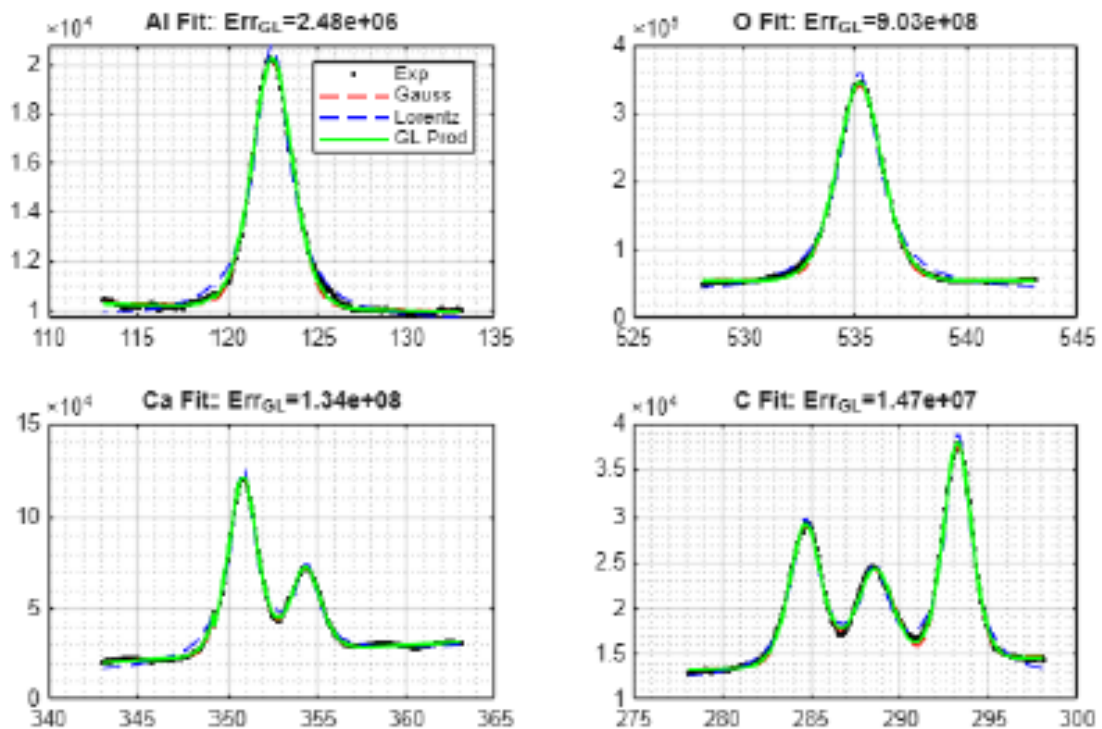
The process begins by importing the raw experimental datasets provided for the study. An initial graphical representation of the survey and specific regional spectra was rendered. This step serves as a critical quality control measure to verify that our imported data aligns perfectly with the reference spectra presented in the official laboratory guidelines.



2. Multi-Model Fitting Optimization (Section 1 & 2)

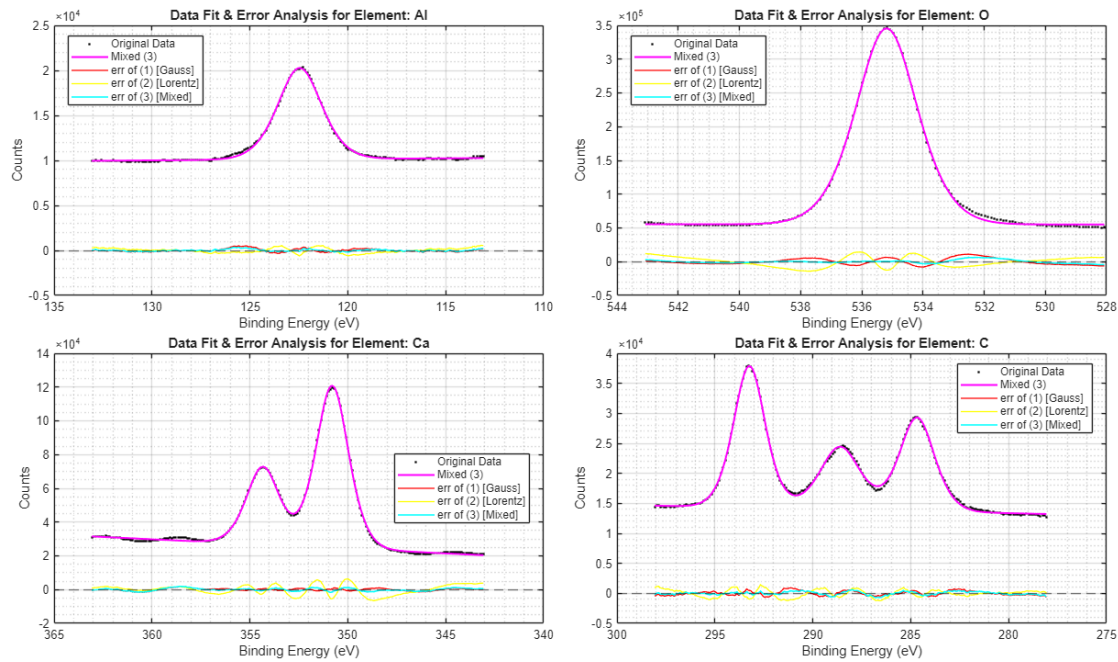
Section 1 requires performing a mathematical peak fitting procedure on the data using three distinct line-shape models: Gaussian (G), Lorentzian (L), and the mixed Gaussian-Lorentzian Product (GL). While literature establishes beforehand that the mixed GL formulation yields the most accurate physical representation for these core levels—making this exercise a validation rather than an open-ended demonstration—each model was optimized independently.

To ensure algorithmic stability, a unique linear background component was extracted from the peak functions and evaluated externally during summation. This approach prevented the localized collapse of the optimization boundaries. Per the guidelines, the three resulting theoretical profiles were plotted alongside the experimental data points for direct comparison. However, to maintain efficiency, only the optimized parameters belonging to the superior GL model were archived in the core memory loop for subsequent analytical utilization.



3. Error Validation and Composite Analysis (Section 4)

Rather than evaluating the quality of the convergence blindly, all individual residual tracking paths were retained simultaneously inside the optimization loop. The final validation subplots elegantly superimpose the original experimental data, the master GL curve, a zero reference baseline, and all three localized error vectors on the same figure axes using a maximized, full-bleed screen layout. This layout clearly reveals how the mixed product minimizes the structural oscillations around the peak core.



OPTIMIZED GL PARAMETERS TABLE (INDIVIDUAL BACKGROUNDS)						
Element	Peak No	Area A	Position x0	Gamma	Bkg Slope a	Bkg Intercept b
Al	1	59204.43	122.4608	1.9484	-15.2677	11985.59
O	1	1355158.32	535.1885	1.7432	24.9028	41916.77
Ca	1	311140.87	350.7991	1.4522	557.6514	-170960.72
Ca	2	151840.79	354.3378	1.4660	557.6514	-170960.72
C	1	57875.33	284.6783	1.5559	69.3550	-6064.12
C	2	51692.12	288.6093	1.7934	69.3550	-6064.12
C	3	71813.36	293.2414	1.4075	69.3550	-6064.12

4. Chemical Composition and Conversion (Section 3)

Utilizing the archived optimal intensity areas and applying the corresponding cross-section Relative Sensitivity Factors (S), the relative atomic concentrations were calculated. Through a straightforward stoichiometric relation between Carbonate and Calcium intensities, the final chemical conversion fraction was successfully determined:

$$\text{CaO to CaCO}_3 \quad \text{Conversion Fraction} = 77.32\%$$

5. Extra Verification

A parallel analysis has been performed using Excel in order to see if the results are consistent with those obtained using the MATLAB solver. By replicating the peak models and running Excel's Solver, the optimized parameters showed negligible numerical deviation between both platforms. This extra step successfully verifies the accuracy of our custom MATLAB script and solidifies the reliability of the final 77.32% conversion calculation.

Pictures that shows this extra check are the following ones:

